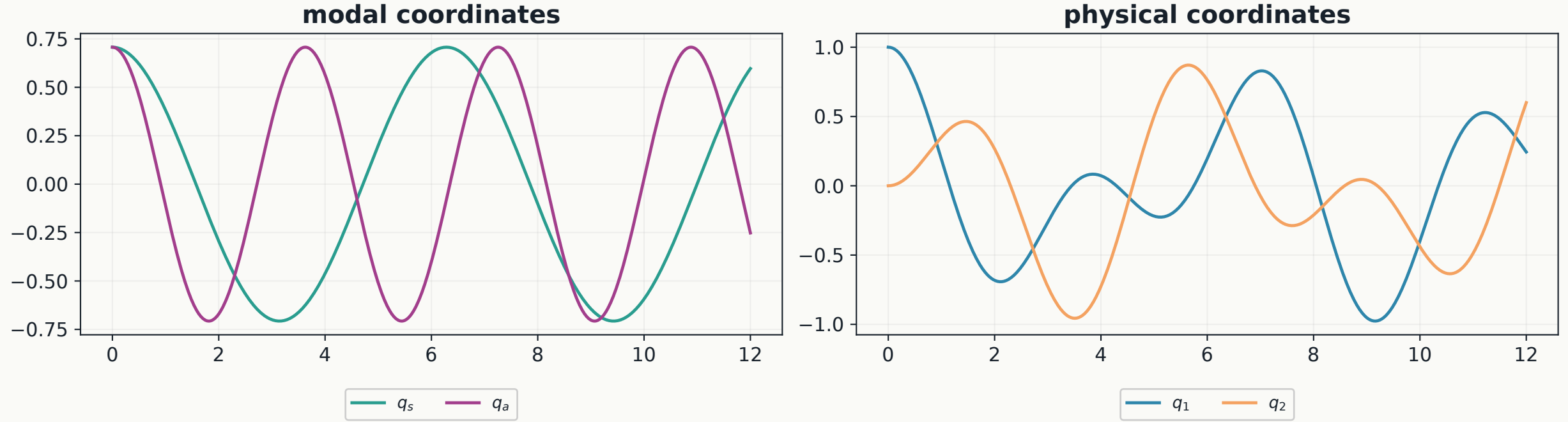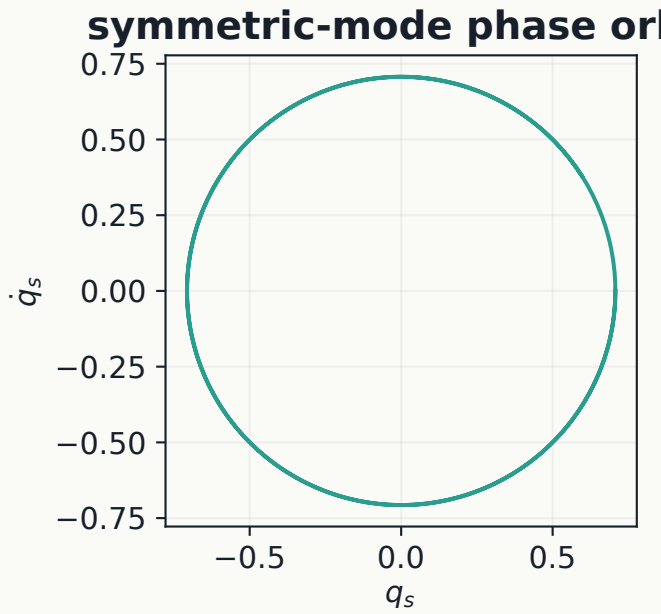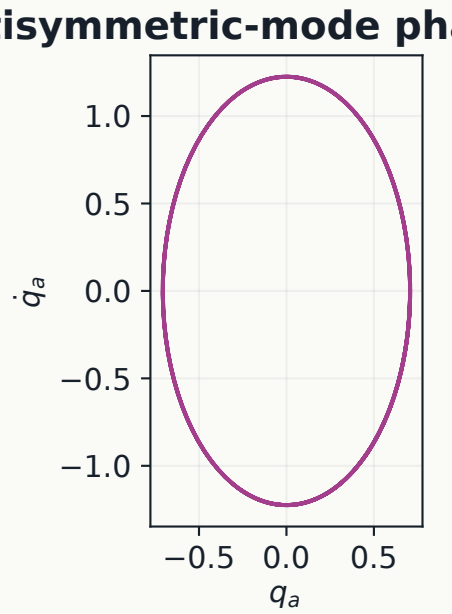Packet evolution is diagonal in modal coordinates and reconstructs coupled physical motion